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AES standard for digital audio — Audio-embedded metadata — Part 3: AAC & HE-AAC

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AES standard for digital audio - Audio-embedded metadata - Part 3: AAC & HE-AAC

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Abstract

AES41 provides for the carriage of audio metadata by embedding it in the audio samples themselves. This tightly associates the metadata with the audio, yet makes it fragile so that changes to the audio will invalidate the metadata. Several metadata sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.

This part describes the format for the data to be transmitted with audio to signal downmix coefficients, loudness and dynamic range control metadata as used in ISO/IEC 14496-3:2009 Subpart 4 (AAC and HE-AAC). A method of carrying this data is described in Part 1 of this Standard.

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Foreword

This foreword is not part of AES41-3-2012 *AES standard for digital audio - Audio-embedded metadata - Part 3: AAC & HE-AAC*

This document describes a set of data that may be conveyed according to the method described in Part 1 of this Standard. It is one part of a multi-part revision of AES41-2009.

As predicted in the foreword to AES41-2000, digital compression techniques now dominate the broadcast television environment. In addition to the problems foreseen relating to cascaded compression, new problems have arisen because of the use of loudness control and surround sound with those digital compression techniques.

Metadata within the compressed audio bit-stream is used to control loudness and the mixing down of multi-channel surround sound to two-channel stereo. These metadata are usually known by terms such as "dialnorm", "prog_ref_level", and "downmix coefficients".

Whilst this might seem unrelated to the original scope of AES41, dealing with bit allocations and scale factors, it is simply another form of data that can affect a later encoding of the audio: this time it is more macroscopic than microscopic.

The metadata is lost when the bit-stream is uncompressed unless provision is made to transport it or store it somewhere. Existing methods rely on non-audio mechanisms to convey the metadata alongside the audio, for example a serial data link like RS-422 and serial digital video SMPTE 259M, or a "chunk" in an audio file (for metadata that does not change).

This part extends AES41 to include a data format for carrying this loudness and downmix metadata with the uncompressed PCM using the same transport mechanism as before. The metadata can therefore be carried in the audio to which it relates.

The draft of this document was developed by a writing group whose primary author was Andrew Mason.

John Grant
Chair, working group SC-02-02
2012-03

Note on normative language

In AES standards documents, sentences containing the word "shall" are requirements for compliance with the document. Sentences containing the verb "should" are strong suggestions (recommendations). Sentences giving permission use the verb "may". Sentences expressing a possibility use the verb "can".

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0 Introduction

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This part describes the format for the data to be transmitted with audio to signal downmix coefficients, loudness and dynamic range control metadata as used in ISO/IEC 14496-3:2009 Subpart 4 (AAC and HE-AAC).

0.1 Rationale for part 3 of this standard.

Unwanted loudness variations in broadcast audio have historically been the source of many audience complaints. Broadcasters have adopted numerous techniques to address the problem - one of which involves indicating the long-term average loudness of the audio.

MPEG-4 AAC and HE-AAC are increasingly being used for broadcast sound services associated with standard- and high-definition television services, in stereo and surround, and the MPEG-4 standard makes provision for metadata to control loudness and downmix. This part of the AES41 standard enable dynamic metadata to be carried to an MPEG-4 audio encoder as part of the PCM audio signal.

1 Scope

This document describes a format for the data to be transmitted to convey a subset of the data in ISO/IEC 14496-3, which included downmix and loudness metadata. This Part assumes that the transmission mechanism according to Part 1 of this Standard is used.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

AES41-1-2012, *AES standard for digital audio - Audio-embedded metadata - Part 1: General*, Audio Engineering Society, New York, NY., US.

ISO/IEC 14496-3:2005(E), *Information technology – Coding of audio-visual objects - Part 3: Audio*. International Electrotechnical Commission, Geneva, Switzerland.

3 Definitions and abbreviations

3.1 Definitions

3.1.1

downmix

The process by which surround sound audio is converted into (two-channel) stereo, usually by adding a fraction of the surround channels and the center channel into the front left and front right channels.

3.1.2

dynamic range control

the process of increasing the loudness of quiet sounds, or of decreasing the loudness of loud sounds, or both, such that the range of signal levels in an audio signal is reduced.

3.1.3

programme reference level

an indication of the average loudness of the audio signal that may be used to effect a normalisation to another target loudness

3.2 Symbols

The mathematical operators used in this standard are similar to those used in the C programming language. The bitwise operators are defined assuming 2's-complement representation of integers. Numbering and counting loops generally begin from zero.

3.2.1 Arithmetic operators

3.2.1.1

+

addition

3.2.1.2

++

increment

3.2.1.3

--

decrement

3.2.1.4

*

multiplication

3.2.1.5

%

modulus operator, defined only for positive numbers

3.2.2 Relational operators

3.2.2.1

<

less than

3.2.2.2

==

equal to

3.2.2.3

!=

not equal to

3.2.3 Bitwise operators

&

AND

3.2.4 Assignment

=

assignment operator

3.3 Abbreviations

The following mnemonics are defined to describe the different data types used in the coded bit stream.

3.3.1

bslbf

bit string, left bit first, where "left" is the order in which bit strings are written in this standard

NOTE Bit strings are written as a string of logic 1's and logic 0's within single quote marks, for example, '1000 0001'. Blanks within a bit string are for ease of reading and have no significance.

3.3.2

simsbf

signed integer, most significant bit first

3.3.3

uimsbf

unsigned integer, most significant bit first

4 ISO/IEC 14496-3:2009 minimal metadata set

4.1 AES41 metadata type

The presence of this data in the stream shall be indicated by **type**=4₁₀, as described in Part 1 of this standard.

4.2 Syntax

Syntax	No. of bits	Mnemonic
<pre> data() { if (type==ISO/IEC 14496-3:2005 minimal metadata) { matrix_mixdown_idx_present matrix_mixdown_idx prog_ref_level_present prog_ref_level reserved, 0 dyn_rng_sgn dyn_rng_ctl } for(p=data_bits%8; p != 0; p--) padding_bit } </pre>	1 2 1 7 1 1 7 4	uimsbf uimsbf uimsbf uimsbf uimsbf uimsbf uimsbf bslbf

4.3 Semantics

The data are a minimal subset of the ISO/IEC 14496-3 metadata, where:

matrix_mixdown_idx_present	indicates whether the following matrix_mixdown_idx is valid:		
	0	not valid, value should be ignored	
	1	valid	
matrix_mixdown_idx	indicates the coefficient “A” as used in the ISO/IEC 14496-3 matrix-mixdown process:		
	0 ₁₀	1/√2	(-3 dB)
	1 ₁₀	1/2	(-6 dB)
	2 ₁₀	1/(2√2)	(-9 dB)
	3 ₁₀	0	(- ∞ dB)
prog_ref_level_present	indicates whether the following prog_ref_level is valid		
	0	not valid, value should be ignored	
	1	valid	
prog_ref_level	indicates program headroom relative to full scale as in ISO/IEC 14496-3, where 0 means the program is full scale and each step is 0,25 dB.		
dyn_rng_sgn	sign bit of the dyn_rng_ctl dynamic range control word		
dyn_rng_ctl	dynamic range control word, quantised in 0,25-dB steps, as in ISO/IEC 14496-3		
padding_bit	indicates bits that are added to bring the total number of bits in the data to a multiple of 8 to match the length field.		

5 Sample alignment of data with respect to audio signal

5.1 General

Frame boundary positions in the decoded PCM audio can be determined from the position of the first bit of the audio coder control data. The relationship between the boundary and the position of this first bit depends on the coding scheme that was last used. Because the anticipated use of this metadata is before MPEG-4 audio coding rather than after, a frame boundary position has not yet been defined. Should it be deemed necessary, later versions of this standard may do so.

5.2 ISO/IEC 14496-3:2009

When the audio coder control data is the ISO/IEC 14496-3:2009 minimal data set, and the audio has not been previously encoded, then the first bit of the audio coder control data shall be embedded in the first audio sample to which it applies. The repetition rate of unchanging audio coder control data may be chosen to suit the requirements for “break-in” at arbitrary time to the audio signal of the application. At one extreme, this would be signaling only once, at the beginning of the stream. More generally, in anticipation of coding, a repetition every 1024 or 960 samples, corresponding to frame lengths in ISO/IEC 14496-3:2009 is recommended.